Outline

1. Sequences, Convergence, Limits
2. Infinite Series
3. Properties of Series, Tests for Convergence
4. Power Series, Taylor & Maclaurin Series

Definition

Power series A power series about $x = a$ is a series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \cdots + c_n(x-a)^n + \cdots .$$

Here x is a variable and c_n's are constants known as the coefficients of the power series.

In particular, we are concerned with a power series about $x = 0$:

$$\sum_{n=0}^{\infty} c_n x^n = c_0 + c_1 x + c_2 x^2 + \cdots + c_n x^n + \cdots .$$

For example, $\sum_{n=0}^{\infty} x^n$ (a special geometric series), $\sum_{n=0}^{\infty} n!(x+3)^n$.

- **Applications of power series to approximations.**

Example

Use the first four terms of the power series for $\dfrac{1}{1-x}$ to estimate

(a) $\dfrac{1}{0.99}$, (b) $\dfrac{1}{1.1}$.

Solution.

(a) Note that $0.99 = 1 - 0.01$. So we use $x = 0.01$ to have

$$\frac{1}{0.99} = \frac{1}{1-0.01} \approx 1 + (0.01) + (0.01)^2 + (0.01)^3 = 1.010101$$

(b) Similarly, since $1.1 = 1 - (-0.1)$, we use $x = -0.1$ to have

$$\frac{1}{1.1} = \frac{1}{1-(-0.1)} \approx 1 + (-0.1) + (-0.1)^2 + (-0.1)^3 = 0.909$$

- **Remarks on convergence of power series**

(a) For each fixed value of x, it is a series of numbers and we may apply the tests for convergence.

(b) It may converge for some values of x or diverge for other values of x.

(c) The sum of a convergent power is a function $f(x)$.

- **Three possible cases for power series** $\sum_{n=0}^{\infty} c_n(x-a)^n$

(i) The power series converges absolutely for all x.

(ii) The power series converges only at one point $x = a$ and diverges elsewhere.

(iii) There is a positive real number R such that the power series converges for $|x-a| < R$ and diverges for $|x-a| > R$.

- **Radius of convergence & Interval of convergence**
  - The number R in case (iii) above is called the radius of convergence of a power series.
  - Also, for case (i), we say the radius of convergence $R = \infty$, while for case (ii), we say $R = 0$.
  - The interval of convergence of the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ is the set of all values for which the power series converges.

The results above show that if R is the radius of convergence, then

(i) For $R = \infty$: the interval of convergence is $(-\infty, \infty)$.

(ii) For $R = 0$: the interval of convergence is $\{a\}$.

(iii) For $R \in (0, \infty)$: the interval of convergence is one of the following

$$(a - R, a + R), \quad [a - R, a + R), \quad (a - R, a + R], \quad [a - R, a + R].$$

For example, the power series $\sum_{n=0}^{\infty} x^n = \frac{1}{1 - x}$ has the radius of convergence $R = 1$ and the interval of convergence $(-1, 1)$.

- **How to finding radius and interval of convergence?**
  - **[Step 1]** Use an appropriate test (geometric series, root test, ratio test, etc.) to find values of x for absolute convergence, and divergence.

    $\Longrightarrow$ This gives us the radius of convergence R.

    There are three possible cases for R:

    (i) The power series converges absolutely for all x: then $R = \infty$.

    (ii) It converges only at one point $x = a$: then $R = 0$.

    (iii) Otherwise, R is a finite positive real number ($0 < R < \infty$). We have a finite interval of convergence.

In cases (i) and (ii), the interval of convergence is respectively $(-\infty, \infty)$ and $\{0\}$

In case (iii), $R \in (0, \infty)$: the power series converges absolutely for $|x - a| < R$ and diverges for $|x - a| > R$.

Then we continue to determine the interval of convergence, that is to check whether the power series converges at $x = a \pm R$.

- **[Step 2]** Determine a convergence at end-points $x = a \pm R$.

  We substitute the values $x = a \pm R$ into the series and apply an appropriate test (comparison test, integral test, alternating series test, ...) to draw the conclusion.

Example

For what values of x does the power series $\sum_{n=1}^{\infty} \frac{x^n}{n2^n}$ converge?

Solution.

- **Step 1.** We use the Ratio Test to determine R.

Let $a_n = \frac{x^n}{n2^n}$. We have

$$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \left|\frac{\frac{x^{n+1}}{(n+1)2^{n+1}}}{\frac{x^n}{n2^n}}\right| = \lim_{n\to\infty} \left|\frac{x^{n+1}}{x^n} \cdot \frac{n}{n+1} \cdot \frac{2^n}{2^{n+1}}\right|$$

$$= \lim_{n\to\infty} \frac{|x|}{2} \frac{n}{n+1} \underbrace{=}_{*} \frac{|x|}{2} \lim_{n\to\infty} \frac{n}{n+1} = \frac{|x|}{2}.$$

Here at $(*)$, x is a constant in the limiting process as $n \to \infty$.

Thus

$$\lim_{n\to\infty}\left|\frac{a_{n+1}}{a_n}\right| = \frac{|x|}{2},$$

and hence by the Ration Test, the given power series

$$\sum_{n=1}^{\infty}\frac{x^n}{n2^n} \quad \begin{cases} \text{converges for } \dfrac{|x|}{2} < 1, \text{ i.e., } |x| < 2 \\ \text{diverges for } \dfrac{|x|}{2} > 1, \text{ i.e., } |x| > 2. \end{cases}$$

Consequently, the radius of convergence is $R = 2$.

- **Step 2.** Consider a convergence at x with $\frac{|x|}{2} = 1$, i.e., $x = \pm 2$.

- At $x = 2$: we have

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n} = \sum_{n=1}^{\infty} \frac{2^n}{n2^n} = \sum_{n=1}^{\infty} \frac{1}{n}$$

is a *harmonic series* which is divergent.

- At $x = -2$: We have

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n} = \sum_{n=1}^{\infty} \frac{(-2)^n}{n2^n} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n}$$

which converges according to alternating series test. (Verify!)

Conclusion: The power series

$$\sum_{n=1}^{\infty} \frac{x^n}{n2^n}$$

converges for all $x \in [-2, 2)$ and diverges otherwise.
The radius of convergence $R = 2$, the interval of convergence $[-2, 2)$.

Note. Alternatively, we can also use the *Root Test* for Step 1:

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{\left|\frac{x^n}{n2^n}\right|} = \lim_{n\to\infty} \frac{|x|}{2\sqrt[n]{n}} = \frac{|x|}{2} \lim_{n\to\infty} \frac{1}{\sqrt[n]{n}} = \frac{|x|}{2},$$

from which all conclusions follow.

Example

For what values of x does the power series $\sum_{n=1}^{\infty} \frac{x^n}{n!}$ converge?

[Answer: $R = \infty$ and the interval of convergence is $(-\infty, \infty)$].

Example

For what values of x does the power series $\sum_{n=1}^{\infty} (n!)(x-1)^n$ converge?

[Answer: $R = 0$ and the interval of convergence is $\{1\}$].

• Term-by-term differentiation

Suppose a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges for $-R < x-a < R$, where $R > 0$. Then it can be differentiated termwise (term-by-term); that is

$$\left(\sum_{n=0}^{\infty} c_n(x-a)^n\right)' = \sum_{n=0}^{\infty} \underbrace{(c_n(x-a)^n)'}_{\text{termwise}} = \sum_{n=1}^{\infty} nc_n(x-a)^{n-1},$$

for $-R < x-a < R$.

On the interval $(a-R, a+R)$, we can also have the higher derivatives of the sum.

Say for the second derivative

$$\begin{aligned}
\left(\sum_{n=0}^{\infty} c_n(x-a)^n\right)'' &= \left(\left(\sum_{n=0}^{\infty} c_n(x-a)^n\right)'\right)' \\
&\underbrace{=}_{\text{termwise}} \left(\sum_{n=1}^{\infty} nc_n(x-a)^{n-1}\right)' \\
&\underbrace{=}_{\text{termwise}} \sum_{n=1}^{\infty}\left(nc_n(x-a)^{n-1}\right)' \\
&= \sum_{n=2}^{\infty} n(n-1)c_n(x-a)^{n-2}.
\end{aligned}$$

• Term-by-term integration

Suppose a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges for $-R < x-a < R$, where $R > 0$. Then it can be integrated termwise (term-by-term); that is

$$\begin{aligned}\int \left(\sum_{n=0}^{\infty} c_n(x-a)^n\right) dx &= \sum_{n=0}^{\infty} \Big(\underbrace{\int c_n(x-a)^n\, dx}_{\text{termwise}}\Big) \\ &= \sum_{n=0}^{\infty} \frac{c_n(x-a)^{n+1}}{n+1} + C,\end{aligned}$$

for $-R < x-a < R$, where C is an arbitrary constant.

Example

Let $f(x) = \dfrac{1}{1-x} = \displaystyle\sum_{n=0}^{\infty} x^n$, with $-1 < x < 1$.

(a) Find a power series of $f'(x)$.

(b) Find a power series of the function $\dfrac{1}{(1-x)^3}$.

Solution. Note that the power series $\displaystyle\sum_{n=0}^{\infty} x^n$ has the radius of convergence $R = 1$, i.e. it converges absolutely for $x \in (-1, 1)$.

(a) Differentiating $f(x)$, we have $f'(x) = \dfrac{1}{(1-x)^2} \quad (|x| < 1)$.

On the other hand, using a power series representation of $f(x)$, we obtain

$$\left(\sum_{n=0}^{\infty} x^n\right)' \underbrace{=}_{\text{termwise}} \sum_{n=1}^{\infty} (x^n)' = \sum_{n=1}^{\infty} n x^{n-1}$$
$$= 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots \quad (-1 < x < 1).$$

Thus we get a power series representation of $f'(x)$ for $-1 < x < 1$:

$$f'(x) = \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = 1 + 2x + 3x^2 + 4x^3 + 5x^4 + \cdots .$$

(b) Next, differentiating again $f'(x)$ in two ways as above, we have $f''(x) = \dfrac{2}{(1-x)^3}$, and also

$$
\begin{aligned}
f''(x) &= \underbrace{\Big(\sum_{n=1}^{\infty} n x^{n-1}\Big)'}_{f'(x)} = \sum_{n=1}^{\infty} \underbrace{\Big(n x^{n-1}\Big)'}_{\text{termwise}} \\
&= \sum_{n=2}^{\infty} n(n-1)x^{n-2} = \underbrace{\sum_{m=0}^{\infty} (m+2)(m+1)x^{m}}_{m=n-2} \\
&= 2 + 6x + 12x^2 + 20x^3 + \cdots + (m+2)(m+1)x^m + \cdots
\end{aligned}
$$

Hence, we have the following power series representation of $\dfrac{1}{(1-x)^3}$:

$$\begin{aligned}\frac{1}{(1-x)^3} &= \frac{1}{2}\sum_{n=0}^{\infty}(n+2)(n+1)x^n = \sum_{n=0}^{\infty}\frac{(n+2)(n+1)}{2}x^n \\ &= 1+3x+6x^2+10x^3+\cdots \quad (-1<x<1).\end{aligned}$$

Example

Determine the power series of $\ln(1+x)$ for $-1 < x < 1$.

Note that $\displaystyle\int \frac{1}{1+x}\,dx = \ln(1+x) + C$.

We have

$$\frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty}(-x)^n, \quad (-1 < x < 1).$$

Integrating both sides of the last identity, we have

$$\int \frac{1}{1+x}\,dx = \int \sum_{n=0}^{\infty}(-x)^n\,dx,$$

which is (via a termwise integration)

$$\begin{aligned}\ln(1+x) &= \sum_{n=0}^{\infty}\Big(\underbrace{\int (-x)^n\,dx}_{\text{termwise}}\Big) = \sum_{n=0}^{\infty}\Big(\int (-1)^n x^n\,dx\Big)\\ &= \sum_{n=0}^{\infty}(-1)^n\frac{x^{n+1}}{n+1}+C = \left(x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots\right)+C,\end{aligned}$$

where C is an arbitrary constant.

Substituting $x=0\in(-1,1)$, we obtain $C=0$, and hence

$$\ln(1+x)=x-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\cdots \quad (-1<x<1).$$

• Summary

We have obtained the following power series representations which are important in the sequel.

- $\dfrac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + \cdots = \displaystyle\sum_{n=0}^{\infty} x^n \quad (|x| < 1)$
- $\dfrac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + \cdots = \displaystyle\sum_{n=0}^{\infty} (-1)^n x^n \quad (|x| < 1)$
- $\ln(1+x) = x - \dfrac{x^2}{2} + \dfrac{x^3}{3} - \dfrac{x^4}{4} + \cdots = \displaystyle\sum_{n=1}^{\infty} (-1)^{n+1} \dfrac{x^n}{n} \quad (|x| < 1)$

• Power series representation

If a function f has a power series representation at $x = a$, that is, if

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \quad (|x-a| < R),$$

then

$$c_n = \frac{f^{(n)}(a)}{n!}, \quad (n \geq 0).$$

This follows from a termwise differentiation, and substitution of $x = a$ to obtain $f^{(n)}(a)$. Indeed, we have the following.

$$f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n \implies f(a) = c_0$$

$$f'(x) = \sum_{n=1}^{\infty} n c_n (x-a)^{n-1} \implies f'(a) = c_1$$

$$f''(x) = \sum_{n=2}^{\infty} n(n-1) c_n (x-a)^{n-2}$$

$$\implies f''(a) = 2c_2 \implies c_2 = \frac{f''(a)}{2}$$

$\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots$

$$f^{(k)}(x) = \sum_{n=k}^{\infty} n(n-1)(n-2)\cdots(n-k+1) c_n (x-a)^{n-k}$$

$$\implies f^{(k)}(a) = k! c_k \implies c_k = \frac{f^{(k)}(a)}{k!}$$

$\ldots \quad \ldots \quad \ldots \quad \ldots \quad \ldots$

Suppose $f(x)$ is a function with all derivatives of all orders throughout some interval containing a as an interior point.

Definition (Taylor series)

The Taylor series generated by $f(x)$ at $x = a$, is the power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots$$

Note. The coefficient of $(x-a)^n$ is $c_n = \dfrac{f^{(n)}(a)}{n!}$.

In particular, when $a = 0$, the series has a special name, a Maclaurin series.

Definition (Maclaurin series ($a = 0$))

The Maclaurin series generated by $f(x)$, is the power series at $a = 0$, i.e.

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots$$

Example

Find the Taylor series generated by $f(x) = \dfrac{1}{x}$ at $a = 3$.

Solution. To determine the coefficient $c_n = \dfrac{f^{(n)}(3)}{n!}$, we evaluate values $f^{(n)}(3)$:

$$f(x) = \frac{1}{x}, \quad f'(x) = \frac{-1}{x^2}, \quad f''(x) = \frac{(-1)(-2)}{x^3} = \frac{(-1)^2 2!}{x^3},$$
$$f'''(x) = \frac{(-1)^3 3!}{x^4}, \quad \ldots, \quad f^{(n)}(x) = \frac{(-1)^n n!}{x^{n+1}}, \quad \ldots$$

Thus $c_n = \dfrac{f^{(n)}(3)}{n!} = \dfrac{(-1)^n n!}{n! 3^{n+1}} = \dfrac{(-1)^n}{3^{n+1}}$.

Therefore, the Taylor series generated by $f(x) = \dfrac{1}{x}$ at $a = 3$ is

$$f(3) + \frac{f'(3)}{1!}(x-3) + \frac{f''(3)}{2!}(x-3)^2 + \cdots + \frac{f^{(n)}(3)}{n!}(x-3)^n + \cdots$$
$$= \frac{1}{3} - \frac{1}{3^2}(x-3) + \frac{1}{3^3}(x-3)^2 + \cdots + \frac{(-1)^n}{3^{n+1}}(x-3)^n + \cdots,$$

which is a geometric series with the first term $\dfrac{1}{3}$ and the common ratio $r = \dfrac{-(x-3)}{3}$.

Note. Since the Taylor series generated by $f(x) = \dfrac{1}{x}$ at $a = 3$ is a *geometric series* with the first term $\dfrac{1}{3}$ and the common ratio $r = \dfrac{-(x-3)}{3}$, we have

- the series converges absolutely for $|r| < 1$, i.e. $|x - 3| < 3$,
- the sum of the series is

$$\frac{\dfrac{1}{3}}{1 - \left(\dfrac{-(x-3)}{3}\right)} = \frac{1}{x} = f(x).$$

• Some important remarks

- In general, the Taylor series generated by a function may or may not be $f(x)$.
- However, the theorem (about power series representation) says that if f is representable as a power series at a, then the power series is its Taylor series.
- Many functions have power series representation, e.g., $\frac{1}{1+x}$, $\ln(1+x)$, e^x, $\sin x$, $(1+x)^\alpha$.
- However, there exist functions that are differentiable but has no power series representation.

Definition (Taylor polynomials)

The Taylor polynomial of degree n of f, denoted by $T_n(x)$, is defined as follows:

$$\begin{aligned} T_n(x) &= f(a) + \frac{f'(a)}{1!}(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots \\ &+ \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n. \end{aligned}$$

The coefficient of $(x-a)^k$ is $c_k = \dfrac{f^{(k)}(a)}{k!} \quad (k = 0, 1, 2, \ldots, n)$.

We have the following notes on Taylor polynomials.

(a) The k-derivatives of $T_n(x)$ and $f(x)$ have the same values at $x = a$, for $1 \leq k \leq n$.

(b) Recall that the linearization of $f(x)$ at a is

$$L(x) = f(a) + f'(a)(x - a),$$

which is $T_1(x)$.

(c) Like linearization, we may use Taylor polynomial $T_n(x)$ to approximate $f(x)$ for x close to a.

When $n = 2$, the polynomial $T_2(x)$ is the quadratic approximation of f for x close to a.

A formula of Taylor remainder

Suppose a function f has derivatives of all orders in an open interval I containing a. Then for each $x \in I$ and each positive integer n, we have

$$f(x) - T_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1},$$

where c is some value between a and x.

The expression $\boxed{R_n = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}}$ is called the Taylor remainder of order n or the error term for the approximation of f by $T_n(x)$.

Thus we have

$$f(x) = \overbrace{f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n}^{T_n(x)} + \underbrace{\frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}}_{R_n(x)},$$

where c is in between a and x.

• Application 1 - Convergence

Since $f(x) = T_n(x) + R_n(x)$, we wish to know whether

$$f(x) = \lim_{n\to\infty} T_n(x).$$

We have the following result.

Theorem

If $\lim_{n\to\infty} R_n(x) = 0$ *for all* x, *then* $f(x) = \lim_{n\to\infty} T_n(x)$, *that is the Taylor series generated by* f *converges to* f.

In other words, under the condition $\lim_{n\to\infty} R_n(x) = 0$ *for all* x, f *is equal to the sum of its Taylor series:*

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Example

Use the Taylor remainder to prove that $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$.

Solution. Note that $f(x) = f'(x) = f''(x) = \cdots f^{(n)}(x) = \cdots = e^x$, and so derivatives of f at $x = 0$ are

$$f(0) = f'(0) = f''(0) = \cdots = f^{(n)}(0) = \cdots = 1, \quad \text{for all } n.$$

Hence the Maclaurin series of $f(x) = e^x$ is

$$1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!},$$

and its Taylor remainder is

$$R_n(x) = \frac{e^c \cdot x^{n+1}}{(n+1)!}, \quad \text{where } c \text{ is in between 0 and } x.$$

To show $\lim_{n\to\infty} R_n(x) = 0$, we consider two cases:

- If $x \leq 0$, then $0 < e^c \leq 1$, and we have

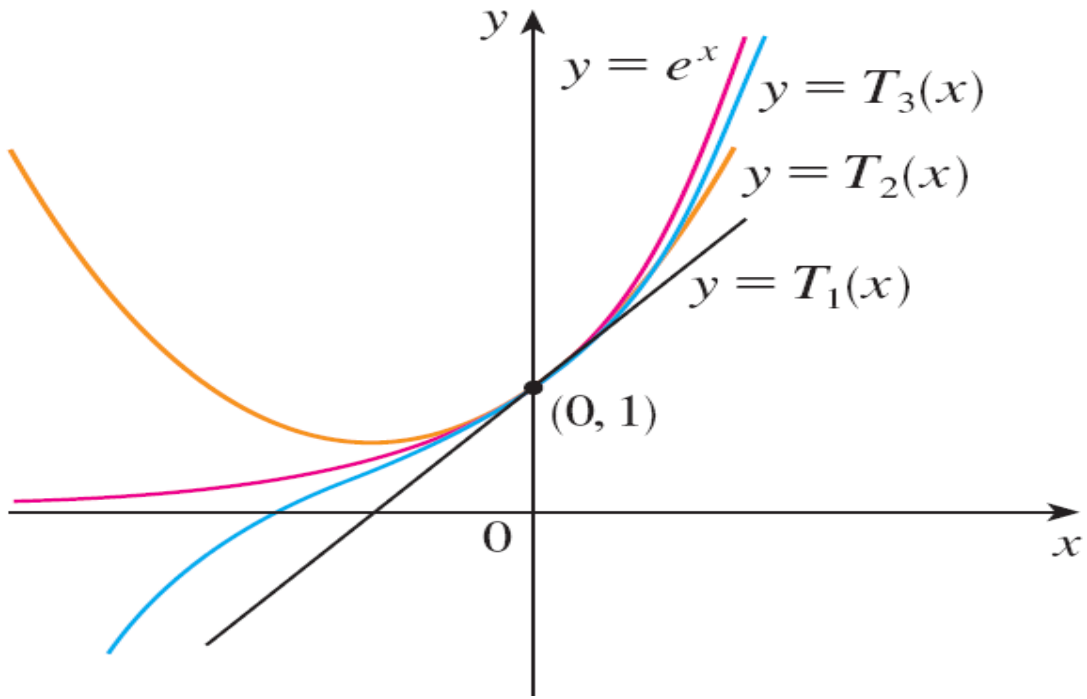
$$|R_n(x)| = \frac{e^c \cdot |x|^{n+1}}{(n+1)!} \leq \frac{|x|^{n+1}}{(n+1)!} \to 0, \quad \text{as } n \to \infty.$$

- If $x > 0$, then since $0 < c < x$ we have $0 < e^c < e^x$, and hence

$$|R_n(x)| = \frac{e^c \cdot |x|^{n+1}}{(n+1)!} \leq e^x \frac{|x|^{n+1}}{(n+1)!} \to 0, \quad \text{as } n \to \infty.$$

So by Squeeze Theorem, $\lim_{n\to\infty} R_n(x) = 0$, and we conclude that e^x is equal to its Maclaurin series for every x, i.e.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + \cdots.$$


y
y = e^x
y = T_3(x)
y = T_2(x)
y = T_1(x)
(0, 1)
0
x

• Application 2 - Error estimation

From $f(x) = T_n(x) + R_n(x)$, we have

$$|f(x) - T_n(x)| = |R_n(x)|, \quad \text{an error},$$

where

$$R_n = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}$$

is the Taylor remainder.

So when we approximate a function $f(x)$ by its Taylor polynomial $T_n(x)$, the Taylor's remainder R_n provides the error in the estimation.

Example

Estimate the error when $T_4(x)$ is used to estimate the value of e^x at $x = 0.5$.

Solution. The approximation of e^x by $T_4(x)$ is

$$e^x \approx T_4(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!},$$

and hence with $x = 0.5$ we have

$$e^{0.5} \approx T_4(0.5) \approx 1.6484.$$

The error when $T_4(0.5)$ is used to estimate $f(0.5) = e^{0.5}$ is $R_4(0.5)$.

We have

$$\begin{aligned} |R_4(0.5)| &= \left| \frac{f^{(5)}(c)}{5!}(0.5-0)^5 \right| \quad (c \in (0, 0.5)) \\ &= \frac{e^c \cdot (0.5)^5}{5!} = 0.00026 \cdot e^c < 0.00026 \cdot e^{0.5} \\ &< 0.00026 \cdot 2 = 0.00052, \end{aligned}$$

because $1 = e^0 < e^c < e^{0.5} < 4^{0.5} = 2$.

• Application 3 - Number of terms for estimation

We can also use the Taylor formula for the remainder to determine number of terms required such that $f(x) \approx T_n(x)$ within a desired error.

Example

Find the number n for approximating e by Taylor polynomial $T_n(1)$ so that the error is at most 10^{-5}.

We have

$$e^x = \underbrace{1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!}}_{T_n(x)} + R_n(x),$$

where $R_n(x) = \dfrac{f^{(n+1}(c) \cdot x^{n+1}}{(n+1)!} = \dfrac{e^c x^{n+1}}{(n+1)!}$, c between 0 and x.

We estimate $e = e^1$ by its Taylor polynomial $T_n(1)$, where

$$T_n(1) = 1 + 1 + \frac{1^2}{2!} + \frac{1^3}{3!} + \cdots + \frac{1^n}{n!} = 2 + \frac{1}{2!} + \frac{1}{3!} + \cdots + \frac{1}{n!},$$

and the error is $R_n(1)$.

We have

$$R_n(1) = \frac{e^c \cdot 1^{n+1}}{(n+1)!} = \frac{e^c}{(n+1)!}, \quad \text{for some } c \in (0,1).$$

Since $e^c < e < 3$, we have

$$|R_n(1)| < \frac{3}{(n+1)!}.$$

By evaluating $\frac{3}{(n+1)!}$ for some n's:

$$\frac{3}{(7+1)!} < 7.4 \times 10^{-5}, \quad \frac{3}{(8+1)!} < 8.3 \times 10^{-6} < 10^{-5},$$

we conclude that

$$|R_n(1)| < 10^{-5}, \quad \text{if } n \geq 8.$$

Thus using $T_8(1)$ to approximate e, the error is at most 10^{-5}.

END OF CHAPTER 6